

Beijing Summer School

Appendix 3: Solving Differential Equations Numerically

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“The question of whether a computer can think is no more interesting than the question of whether a submarine can swim.”

–Edsger W. Dijkstra

Goals

- “Traditional” approaches to solving differential equations (especially finite difference)
- Introduce neural networks
- Outline algorithms for training neural networks
- Discuss different software packages
- Apply algorithms to solving differential equations
- Aim: everyone solves a differential equation with deep learning

References on Traditional Techniques for Solving PDEs

- Finite differences: [LeVeque, 2007], [Presse et al., 1988]
- Spectral method: [Pozrikidis, 2005], [Trefethen, 2001]
- Other references:
 - Differential equation theory: Evans (2001)
 - Technique comparisons: [Pichler et al., 2013], [Trefethen, 1996]
 - Introductory texts: [Holmes, 2007], [Iserles, 2008]
 - Upwind scheme in macroeconomic models: [Candler, 2001]

Suggested References on Deep Learning

- [Goodfellow et al., 2016]: “Deep Learning”
 - Typically recommended as a mathematical reference on deep learning
 - Doesn’t provide coding examples
- [Zhang et al., 2021]: “Dive Into Deep Learning”
 - Builds python code from “from scratch”, which helps “de-mystify” NNs
 - Also provides sample code in TensorFlow, PyTorch and Gluon
- [Géron, 2019]: “Hands-on Machine Learning”. 2nd Edition.
 - 2nd Edition introduces Neural Networks using the Keras interface to TensorFlow 2 (one of the main deep learning systems)
 - Doesn’t cover mathematical details

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Population model

- Suppose that agents have an individual state evolving according to $dX_t^i = \sqrt{2}\sigma dW_t^i$.
- Agents exit when they reach $x \in (x_L, x_H)$.
- New agents arrive at x at rate $\lambda_b(x) = f(x)$.

What is the KFE for this population?

KFE

- Recall: the generic KFE (for Brownian rather than Poisson shocks) is:

$$\partial_t g_t(x) = \lambda_b(x) - \lambda_k(x)g_t(x) - \partial_x[\mu_t(x)g_t(x)] + \frac{1}{2}\partial_{xx}[\sigma_t^2(x)g_t(x)]$$

- So for individual evolution $dX_t^i = \sqrt{2}\sigma dW_t^i$ and absorbing boundaries the KFE is:
(In maths, this is called the heat equation or diffusion equation)

$$\partial_t g_t(x) = f(x) + \sigma^2 \partial_{xx} g_t(x), \quad g(x_L) = 0, \quad g(x_H) = 0$$

- We start by solving for the steady state heat equation for $\sigma = 1$ (which is also the 1D Poisson equation). This is a *boundary value problem*.

$$0 = f(x) + \partial_{xx} g_t(x), \quad \text{s.t.} \quad g(x_L) = 0, \quad g(x_H) = 0$$

- We then solve the full, time varying heat equation. This is an *initial value problem*.

$$\partial_t g_t(x) = f(x) + \sigma^2 \partial_{xx} g_t(x), \quad \text{s.t.} \quad \text{boundary conditions} + \text{initial value conditions}$$

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Example 1: Steady State KFE With Exponential Entry Rate

- **Q. Problem:** Consider the differential equation (steady state diffusion / 1D Poisson)

$$0 = f(x) + \partial_{xx}g(x), \quad \text{where } x \in (x_L, x_H), \quad f(x) = e^x$$

subject to the boundary conditions (referred to as “Dirichlet” boundary conditions):

$$g(x_L) = g_L = 0, \qquad g(x_H) = g_H = 0$$

- **A. Analytic solution:** Integrating twice (or using Mathematica) gives general solution:

$$g(x) = -e^x + Cx + D$$

Substituting in the boundary condition solving for C and D gives:

$$C = \frac{g_H - g_L + (e^{x_H} - e^{x_L})}{x_H - x_L} \qquad D = g_H + e^{x_H} - Cx_H$$

How can we solve this numerically with finite difference?

Discretize state space

- Discretize interval $[x_L, x_H]$ into N equally spaced grid points:

$$\{x_1, x_2, \dots, x_{N-1}, x_N\}, \quad x_1 = x_L, \quad x_N = x_H$$

- Distance between grid points (mesh width) is:

$$h = \frac{x_n - x_1}{N - 1}$$

- Discretize $g(x)$ on $\{x_1, x_2, \dots, x_{N-1}, x_N\}$. Denote the discretized values of g by:

$$\{G_1, \dots, G_N\} = \{g(x_1), \dots, g(x_N)\}$$

Finite Difference Approximation to $\partial_x g(x)$

- We only need to approximate $\partial_{xx}g(x)$, but let's consider $\partial_x g(x)$ now anyway
- **Forward difference** at grid point x_j is: (1st order accurate; error $o(h)$)

$$\hat{\partial}_{F,j}g(x) := \frac{G_{j+1} - G_j}{x_{j+1} - x_j} = \frac{G_{j+1} - G_j}{h}, \quad 1 \leq j \leq N - 1$$

- **Backward difference** at grid point x_j is: (1st order accurate; error $o(h)$)

$$\hat{\partial}_{B,j}g(x) := \frac{G_j - G_{j-1}}{x_j - x_{j-1}} = \frac{G_j - G_{j-1}}{h}, \quad 2 \leq j \leq N$$

- See [LeVeque \(2007\)](#), Chp 1 for derivation from Taylor series.

Finite Difference Approximation to $\partial_{xx}g(x)$

- **Centered approximation** at grid point x_j is: (4th order accurate; error $o(h^2)$)

$$\hat{\partial}_{C,j}^2 g(x) = \hat{\partial}_{F,j} \hat{\partial}_{B,j} g(x) = \frac{G_{j+1} - 2G_j + G_{j-1}}{h^2}, \quad 2 \leq j \leq N-1$$

- See [LeVeque \(2007\)](#), Chp 1 for derivation from Taylor series.

Finite Difference Approximation

- Discretized DE becomes a system of algebraic equations:

$$G_1 = g_L, \quad j = 1$$

$$\frac{1}{h^2}(G_{j-1} - 2G_j + G_{j+1}) = -f(x_j), \quad 2 \leq j \leq N-1$$

$$G_N = g_H, \quad j = N$$

- Alternative: Stack equations 1 to N (all equations) into matrix form:

$$\underbrace{\frac{1}{h^2} \begin{bmatrix} h^2 & 0 & 0 & 0 & 0 & \dots & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & \dots & 0 & 0 \\ & & & \ddots & \ddots & \ddots & & \\ 0 & 0 & \dots & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & \dots & 0 & 0 & 0 & 0 & h^2 \end{bmatrix}}_{:=A} \underbrace{\begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ \vdots \\ G_{N-2} \\ G_{N-1} \\ G_N \end{bmatrix}}_{:=G} = \underbrace{\begin{bmatrix} g_L \\ -f(x_2) \\ -f(x_3) \\ \vdots \\ -f(x_{N-2}) \\ -f(x_{N-1}) \\ g_H \end{bmatrix}}_{:=F}$$

- Can solve: $AG = F \Rightarrow G = A^{-1}F$

Alternative Matrix Formation (Without Boundaries)

- Discretized DE becomes a system of algebraic equations:

$$G_1 = g_L, \quad j = 1$$

$$\frac{1}{h^2}(G_{j-1} - 2G_j + G_{j+1}) = -f(x_j), \quad 2 \leq j \leq N-1$$

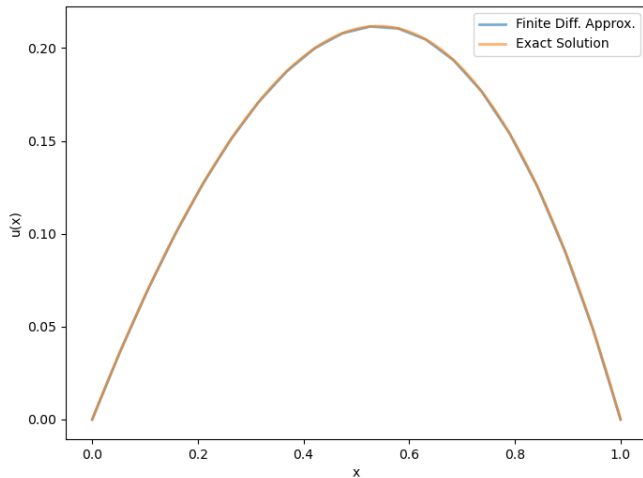
$$G_N = g_H, \quad j = N$$

- Sub in G_1 and G_N and stack equations 2 to $N-1$ into matrix form:

$$\underbrace{\frac{1}{h^2} \begin{bmatrix} -2 & 1 & 0 & 0 & \dots & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 \\ & & \ddots & \ddots & \ddots & \\ 0 & \dots & 0 & 1 & -2 & 1 \\ 0 & \dots & 0 & 0 & 1 & -2 \end{bmatrix}}_{:=A} \underbrace{\begin{bmatrix} G_2 \\ G_3 \\ G_4 \\ \vdots \\ G_{N-2} \\ G_{N-1} \end{bmatrix}}_{:=G} = \underbrace{\begin{bmatrix} -f(x_2) - g_L/h^2 \\ -f(x_3) \\ -f(x_4) \\ \vdots \\ -f(x_{N-2}) \\ -f(x_{N-1}) - g_H/h^2 \end{bmatrix}}_{:=F}$$

- Can solve: $AG = F \Rightarrow G = A^{-1}F$

Sample Code and Solution



Julia/FD_Ex01_PoisEqnSolver_DirBC.jl

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Example 2: Neumann Boundary Conditions

- Consider diffusion equation:

$$0 = f(x) + \partial_{xx}g(x), \quad x \in (x_L, x_H)$$

- Subject to the “mixed Dirichlet and Neumann” boundary conditions:

$$\partial_x g(x_L) = 0$$

$$g(x_H) = 0$$

- and the forcing term:

$$f(x) = e^x$$

- Q. What population model would this correspond to?

Analytical Solution

- Integrate twice and apply boundary conditions.
- This gives the solution:

$$g(x) = -e^x + Cx + D$$

- where

$$C = g_L + e^{x_L}$$

$$D = g_H + e^{x_H} - Cx_H$$

Finite Difference Approximation

- Approximate $g_x(x_L)$ using forward difference (because at lower boundary):

$$\frac{G_2 - G_1}{h} = g_L$$

Stack other equations the same way; gives matrix form:

$$\underbrace{\frac{1}{h^2} \begin{bmatrix} -h & h & 0 & 0 & \dots & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 \\ 0 & 0 & 1 & -2 & \dots & 0 \\ & & \ddots & \ddots & \ddots & \\ 0 & \dots & 1 & -1 & 1 & 0 \\ 0 & \dots & 0 & 1 & -2 & 1 \\ 0 & \dots & 0 & 0 & 0 & h^2 \end{bmatrix}}_{:=A} \underbrace{\begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ \vdots \\ G_{N-2} \\ G_{N-1} \\ G_N \end{bmatrix}}_{:=G} = \underbrace{\begin{bmatrix} g_L \\ -f(x_2) \\ -f(x_3) \\ -f(x_4) \\ \vdots \\ -f(x_{N-2}) \\ -f(x_{N-1}) \\ g_H \end{bmatrix}}_{:=F}$$

Can solve: $G = A^{-1}F$

Finite Difference Approximation (Version 2)

- Introduce G_0 , then approximate $\partial_{xx}g(x_L)$ and $\partial_xg(x_L)$ using central difference:

$$\left. \begin{aligned} \frac{1}{h^2}(G_0 - 2G_1 + G_2) &= -f(x_1), & j=0 \\ \frac{1}{2h}(G_2 - G_0) &= g_L, & j=1 \end{aligned} \right\} \Rightarrow \frac{1}{h}(-G_1 + G_2) = g_L - \frac{h}{2}f(x_1)$$

- Stack equations 0 to N into matrix form and solve: $G = A^{-1}F$:

$$\underbrace{\frac{1}{h^2} \begin{bmatrix} -h & h & 0 & 0 & \dots & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 \\ 0 & 0 & 1 & -2 & \dots & 0 \\ & & \ddots & \ddots & \ddots & \\ 0 & \dots & 1 & -2 & 1 & 0 \\ 0 & \dots & 0 & 1 & -2 & 1 \\ 0 & \dots & 0 & 0 & 0 & h^2 \end{bmatrix}}_{:=A} \underbrace{\begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ \vdots \\ G_{N-2} \\ G_{N-1} \\ G_N \end{bmatrix}}_{:=G} = \underbrace{\begin{bmatrix} g_L - \frac{h}{2}f(x_0) \\ -f(x_2) \\ -f(x_3) \\ -f(x_4) \\ \vdots \\ -f(x_{N-2}) \\ -f(x_{N-1}) \\ g_H \end{bmatrix}}_{:=F}$$

Finite Difference Approximation (Version 3)

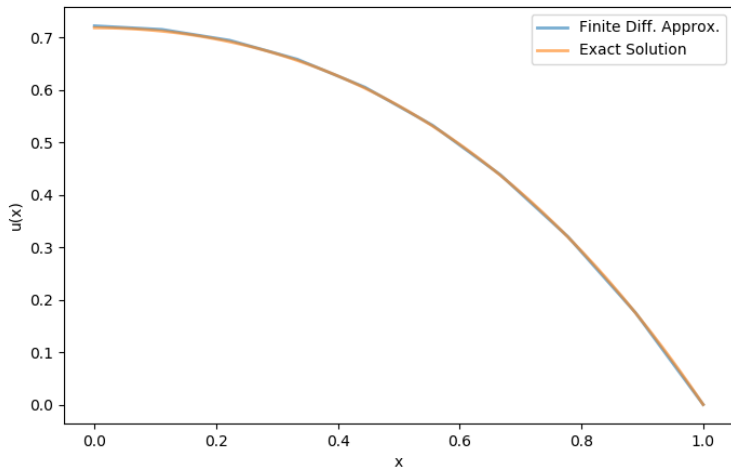
- Use a right one sided approximation accurate to high order $o(h^3)$ (can check this is $o(h^3)$ using Taylor approximation, see Example 1.2 in LeVeque (2007)):

$$(1/h) ((3/2)G_1 - 2G_2 + (1/2)G_3)$$

- Stack equations 0 to N into matrix form and solve for $G = A^{-1}F$:

$$\underbrace{\frac{1}{h^2} \begin{bmatrix} -3h/2 & -2h & h/2 & 0 & \dots & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 \\ 0 & 0 & 1 & -2 & \dots & 0 \\ & & \ddots & \ddots & \ddots & \\ 0 & \dots & 1 & -2 & 1 & 0 \\ 0 & \dots & 0 & 1 & -2 & 1 \\ 0 & \dots & 0 & 0 & 0 & h^2 \end{bmatrix}}_{:=A} \underbrace{\begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ \vdots \\ G_{N-2} \\ G_{N-1} \\ G_N \end{bmatrix}}_{:=G} = \underbrace{\begin{bmatrix} g_L \\ -f(x_2) \\ -f(x_3) \\ -f(x_4) \\ \vdots \\ -f(x_{N-2}) \\ -f(x_{N-1}) \\ g_H \end{bmatrix}}_{:=F}$$

Sample Code and Solution



Julia/FD_Ex02_PoisEqnSolver_FD_NeuBC.jl

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Initial Value Problem

- Consider the one-dimensional *Initial Value Problem* takes the form:
(Note the notation change from g_t to $g(t)$).

$$\partial_t g(t) = f(g(t)), \quad t > 0$$

- subject to the initial condition:

$$g(0) = \psi$$

- NOTE: We can reduce higher order equations to a system of first order equations

Analytical Solution

- Discretise time dimension with uniform time step k :

$$\{t_0, t_1, t_2, t_3, \dots, t_n, \dots\}, \quad t_0 = 0, \quad t_n = nk$$

- Discretise $g(t)$: (using the notation $G^n = g(t_n)$)

$$\{G^0, G^1, \dots, G^N, \dots\} = \{g(t_0), g(t_1), \dots, g(t_N), \dots\}$$

Basic Approaches

- Initial condition: $G_0 = \psi$. Many different algorithms for iterating forward in time:
- **Forward Euler Method:** uses forward difference leading to an *explicit scheme*:

$$\frac{G^{n+1} - G^n}{k} = f(G^n) \quad \Rightarrow \quad G^{n+1} = G^n + kf(G^n), \quad n = 1, 2, \dots$$

- **Backward Euler Method:** uses backward difference leading to an *implicit scheme*:

$$\frac{G^{n+1} - G^n}{k} = f(G^{n+1}) \quad \Rightarrow \quad G^{n+1} = G^n + kf(G^{n+1}), \quad n = 1, 2, \dots$$

- **Trapezoidal Method:** Averages the two Euler methods:

$$\frac{G^{n+1} - G^n}{k} = \frac{f(G^n) + f(G^{n+1})}{2} \quad \Rightarrow \quad G^{n+1} = G^n + k\frac{1}{2} \left(f(G^n) + f(G^{n+1}) \right), \quad n = 1, 2, \dots$$

- Implicit methods may have better stability properties.

Some Other Approaches

- **Runge-Kutta Methods:** propagate a solution over a time step by combining the information from several Euler-style steps. (Slower but typically more robust)
- **Linear Multistep Methods:** Euler and Runge-Kutta methods are *one-step* methods in the sense that G^{n+1} only depends upon G^n . Linear multistep methods allow G^{n+1} to depend a longer history of past steps.
- **Predictor-Corrector Methods:** Extrapolate the solution one period ahead using the current shape of the function and then corrects the extrapolation using derivative information at the new point.

Example 4: Initial Value PDE

- Consider diffusion equation:

$$\partial_t g(x, t) = \partial_{xx} g(x, t) + f(x), \quad x \in (x_L, x_H), \quad t > 0$$

- Subject to the initial condition:

$$g(x, 0) = \eta(x)$$

- Subject to the Dirichlet boundary conditions:

$$g(x_L, t) = g_L(t)$$

$$g(x_H, t) = g_H(t)$$

- With the forcing term:

$$f(x) = e^x$$

Discretise state space

- Discretise space dimension $[x_L, x_H]$ into J equally spaced grid points with width $h = \frac{x_J - x_1}{J-1}$:

$$\{x_1, x_2, \dots, x_j, \dots, x_{J-1}, x_J\}, \quad x_1 = x_L, \quad x_J = x_H$$

- Discretise the time dimension with uniform time step k :

$$\{t_0, t_1, t_2, t_3, \dots, t_n, \dots, t_N\}, \quad t_0 = 0, \quad t_n = nk$$

- Discretise $g(x, t)$: Let $G_j^n := g(x_j, t_n)$

Forward Euler (Explicit Method)

- Discretise system as:

$$\begin{aligned}G_1^{n+1} &= g_L(t_{n+1}), \quad j = 1 \\ \frac{G_j^{n+1} - G_j^n}{k} &= \frac{1}{h^2}(G_{j-1}^n - 2G_j^n + G_{j+1}^n) + f(x_j), \quad 2 \leq j \leq J-1 \\ G_J^{n+1} &= g_H(t_{n+1}), \quad j = J\end{aligned}$$

- For $2 \leq j \leq J-1$, this gives the iteration in the time dimension:

$$G_j^{n+1} = G_j^n + \frac{k}{h^2}(G_{j-1}^n - 2G_j^n + G_{j+1}^n + h^2 f(x_j))$$

Backward Euler (Implicit Method)

- Discretise system as:

$$\begin{aligned}G_1^{n+1} &= g_L(t_{n+1}), \quad j = 1 \\ \frac{G_j^{n+1} - G_j^n}{k} &= \frac{1}{h^2}(G_{j-1}^{n+1} - 2G_j^{n+1} + G_{j+1}^{n+1}) + f(x_j), \quad 2 \leq j \leq J-1 \\ G_J^{n+1} &= g_H(t_{n+1}), \quad j = J\end{aligned}$$

- For $2 \leq j \leq J-1$, this gives the iteration in the time dimension:

$$G_j^{n+1} = G_i^n + \frac{k}{h^2}(G_{j-1}^{n+1} - 2G_j^{n+1} + G_{j+1}^{n+1} + h^2 f(x))$$

Crank-Nicolson (“Trapezoidal” Method)

- Approximate the system using (this method is also implicit):

$$\begin{aligned}G_1^{n+1} &= g_L(t_{n+1}), \quad j = 1 \\ \frac{G_j^{n+1} - G_j^n}{k} &= \frac{1}{2}(D^2 G_j^n + D^2 G_j^{n+1}) + f(x_j), \quad 2 \leq j \leq J-1 \\ &= \frac{1}{2h^2}(G_{j-1}^n - 2G_j^n + G_{j+1}^n + G_{j-1}^{n+1} - 2G_j^{n+1} + G_{j+1}^{n+1}) + f(x_j) \\ G_J^{n+1} &= g_H(t_{n+1}), \quad j = J\end{aligned}$$

- For $2 \leq j \leq J-1$, this can be rewritten as:

$$G_j^{n+1} = G_j^n + \frac{k}{2h^2}(G_{j-1}^n - 2G_j^n + G_{j+1}^n + G_{j-1}^{n+1} - 2G_j^{n+1} + G_{j+1}^{n+1} + h^2 f(x_j))$$

- or the following (where $r = k/(2h^2)$):

$$-rG_{j-1}^{n+1} + (1 + 2r)G_j^{n+1} - rG_{j+1}^{n+1} = rG_{j-1}^n + (1 - 2r)G_j^n + rG_{j+1}^n + rh^2 f(x_j)$$

Finite Difference Approximation (Version 1)

- At each step n , we can stack the equations in matrix form as before:

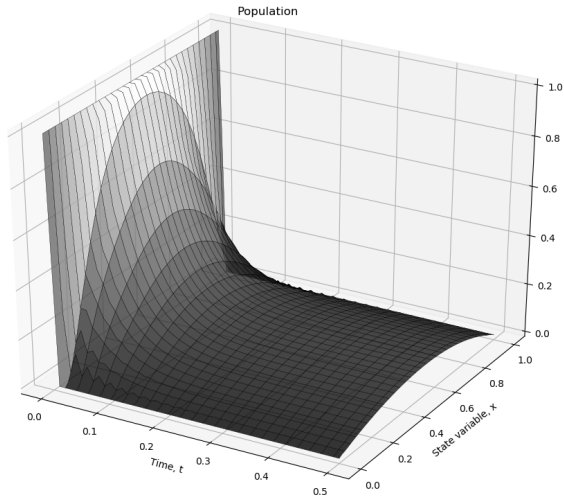
$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \dots & 0 & 0 \\ -r & (1+2r) & -r & 0 & 0 & \dots & 0 & 0 \\ 0 & -r & (1+2r) & -r & 0 & \dots & 0 & 0 \\ & & & \ddots & \ddots & \ddots & & \\ 0 & 0 & \dots & 0 & 0 & -r & (1+2r) & -r \\ 0 & 0 & \dots & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{:=A} \underbrace{\begin{bmatrix} G_1^{n+1} \\ G_2^{n+1} \\ G_3^{n+1} \\ \vdots \\ G_{J-1}^{n+1} \\ G_J^{n+1} \end{bmatrix}}_{:=G^{n+1}} \\
 = \underbrace{\begin{bmatrix} g_L^{n+1} \\ rG_1^n + (1-2r)G_2^n + rG_3^n + rh^2f(x_2) \\ rG_2^n + (1-2r)G_3^n + rG_4^n + rh^2f(x_3) \\ \vdots \\ rG_{J-2}^n + (1-2r)G_{J-1}^n + rG_J^n + rh^2f(x_{J-1}) \\ u_H^{n+1} \end{bmatrix}}_{:=F(G^n)}$$

Algorithm

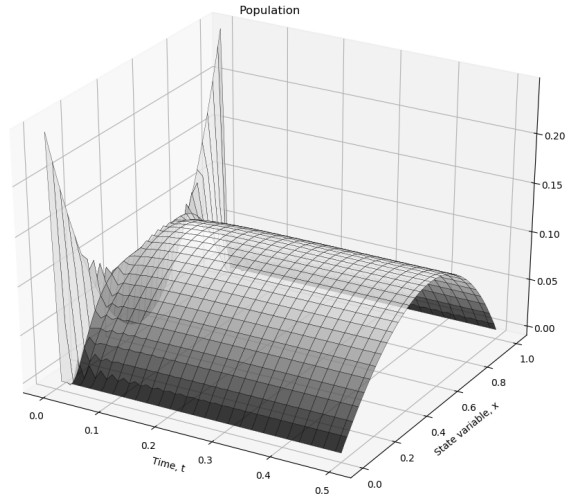
- For $n = 0$, start with the initial density $G^0 = (G_1^0, \dots, G_J^0)$
- For each $n > 0$, solve:

$$G^{n+1} = A^{-1}F(G^n)$$

Sample Code ($g(x, 0) = \eta(x) = 1$)



Sample Code ($g(x, 0) = \eta(x) = (x - 0.5)^2$)



Q: What are some potential problems with this technique?

Q: Could we use this technique for a differential equation with 100 dimensions?

Q/A

Q: What are some potential problems with this technique?

- **A.** We are approximating derivatives with slopes between grid points.
- If the problem has high curvature, then we need a very fine grid, which is computationally hard
- Alternative is to project V onto basis functions rather than grid points.

Q: Could we use this technique for a differential equation with 100 dimensions?

- **A.** No chance!
- If we have N grid points in very dimension, then total matrix dimension is N^{100} .
- Physics literature has turned to deep learning to solve high dimensional problems.
- The idea is to represent V by a neural network and then train parameters so neural network solves the differential equation.

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Multi-Layer Perceptrons (MLPs)

- An MLP attempts to approximate some function $\mathbf{y} = f^*(\mathbf{x})$, where $\mathbf{x}$ and $\mathbf{y}$ are vectors of dimension p and q
- Approximation uses a “network” structure:
 - Approximates f^* by a mapping $y = f(\mathbf{x}, \theta)$, with parameters θ
 - The mapping, f , is typically represented by composing many different functions in a chain (i.e. a *network structure*)
- Other names: *feedforward* or *deep feedforward neural networks*
- Ultimately, we will “learn” the parameters, θ , that give “best” approximation

General MLP Structure

- In general, an MLP neural network approximation takes the form:

$$\mathbf{h}^{(1)} = \phi^{(1)}(W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \quad \dots \text{Hidden layer 1}$$

$$\mathbf{h}^{(2)} = \phi^{(2)}(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \quad \dots \text{Hidden layer 2}$$

$$\vdots$$

$$\mathbf{h}^{(H)} = \phi^{(H)}(W^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) \quad \dots \text{Hidden layer H}$$

$$\mathbf{o} = W^{(H+1)}\mathbf{h}^{(H)} + \mathbf{b}^{(H+1)} \quad \dots \text{Output layer}$$

$$\mathbf{y} = \sigma(\mathbf{o}) \quad \dots \text{Output}$$

- Terminology:
 - H : is the number of *hidden layers*
 - Length of vector $\mathbf{h}^{(i)}$: number of *neurons* in hidden layer i
 - $\phi^{(i)}$: is the *activation function* for hidden layer i
 - σ : is the *activation function* for the output layer
 - $\theta = (W^1, \dots, W^{(H+1)}, b^{(1)}, \dots, b^{(H+1)})$ are the *parameters*

Ex 1: No Hidden Layers, Linear Network

- Suppose that $H = 0$ (no hidden layers); $\mathbf{x} \in \mathbb{R}^4$; $\mathbf{y} \in \mathbb{R}^3$
- In this case, the neural network is just a linear combination:
 - In matrix notation: $\mathbf{o} = W\mathbf{x} + \mathbf{b}$ and $\mathbf{y} = \mathbf{o}$

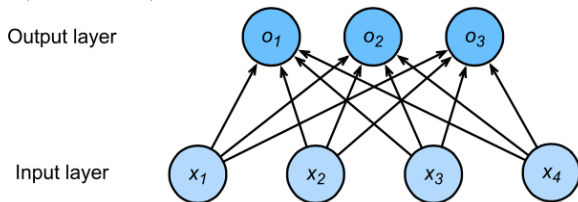
- Without matrix notation:

$$o_1 = x_1W_{11} + x_2W_{12} + x_3W_{13} + x_4W_{14} + b_1, \quad y_1 = o_1$$

$$o_2 = x_1W_{21} + x_2W_{22} + x_3W_{23} + x_4W_{24} + b_2, \quad y_2 = o_2$$

$$o_3 = x_1W_{31} + x_2W_{32} + x_3W_{33} + x_4W_{34} + b_3, \quad y_3 = o_3$$

- Graphical depiction (often used):



- This is called a *dense* neural network because it is “fully connected”

Ex 2: One Hidden Layer; Linear Network

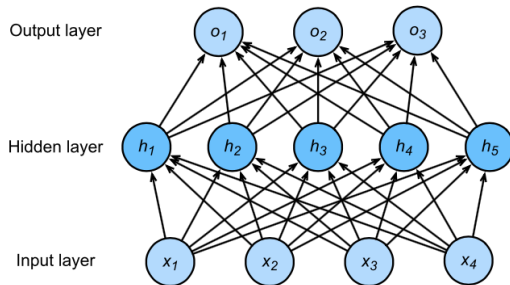
- Suppose $H = 1$ (one hidden layer with 5 neurons); $\mathbf{x} \in \mathbb{R}^4$; $\mathbf{y} \in \mathbb{R}^3$
- Neural network (in matrix notation):

$$\mathbf{h} = W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}, \quad W^{(1)} \text{ is } 5 \times 4, \quad \mathbf{b}^{(1)} \text{ is } 5 \times 1$$

$$\mathbf{o} = W^{(2)}\mathbf{h} + \mathbf{b}^{(2)}, \quad W^{(2)} \text{ is } 3 \times 5, \quad \mathbf{b}^{(2)} \text{ is } 3 \times 1$$

$$\mathbf{y} = \mathbf{o}$$

- Neural network (graphically):



Did the hidden layer add anything?

Adding Linear Layers

- Introducing additional linear layers doesn't add anything
- Why? Because the composition of linear layers is still linear
- Instead, we need to add non-linear activation functions to generate more complicated approximations

Ex 3: One Hidden Layer; Non-Linear Network

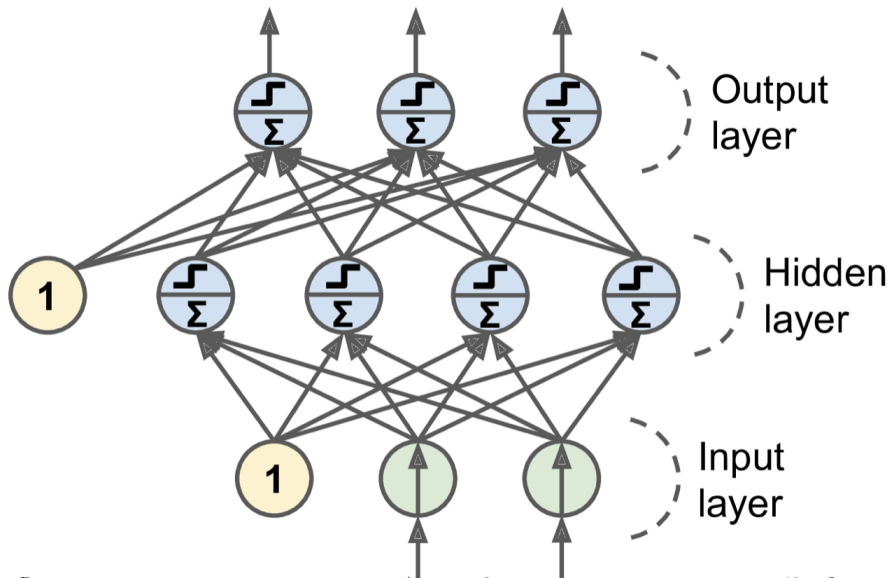
- Suppose $H = 1$ (one hidden layer with 4 neurons); $\mathbf{x} \in \mathbb{R}^2$; $\mathbf{y} \in \mathbb{R}^3$
- Neural network (in matrix notation):

$$\begin{aligned}\mathbf{h} &= \phi(W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}), & W^{(1)} \text{ is } 4 \times 2, & \quad \mathbf{b}^{(1)} \text{ is } 4 \times 1 \\ \mathbf{o} &= W^{(2)}\mathbf{h} + \mathbf{b}^{(2)}, & W^{(2)} \text{ is } 3 \times 4, & \quad \mathbf{b}^{(2)} \text{ is } 3 \times 1 \\ \mathbf{y} &= \mathbf{o}\end{aligned}$$

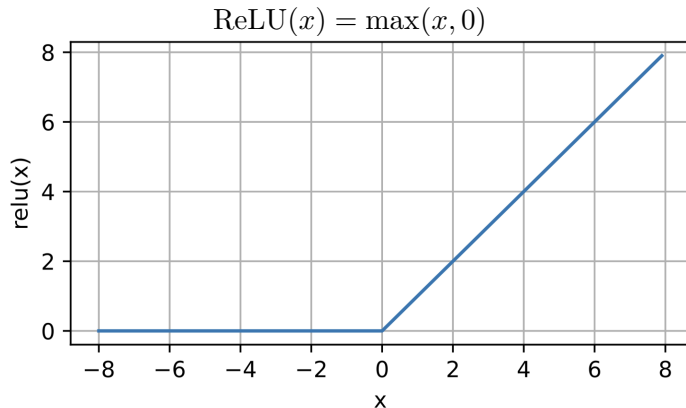
where the non-linear *activation function* ϕ is applied to each element of the vector $W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}$

- Common ϕ functions:
 - Rectified linear unit (ReLU) function
 - Sigmoid function
 - Tanh function

Aside: Geron (2019)'s Graphical Depiction

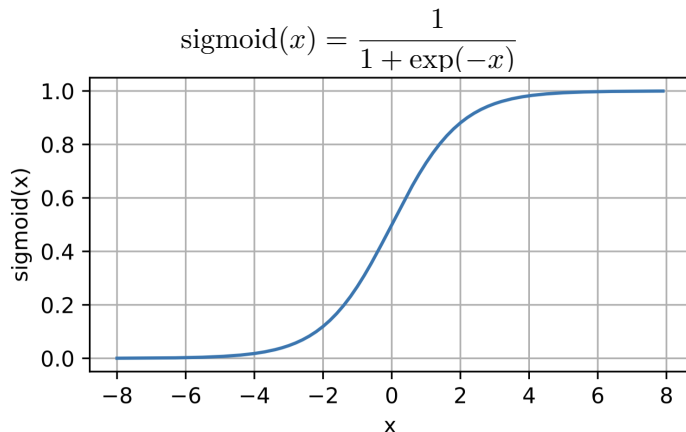


Activation Function: ReLU



- The Rectified Linear Unit (ReLU) function retains only positive elements
- Informally, the ReLU function retains only positive elements

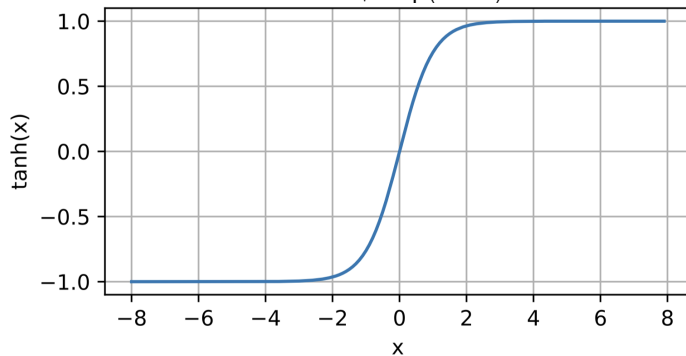
Activation Function: Sigmoid Function



- Transforms (“squashes”) the values into the interval $(0, 1)$
- Helpful if the function f is smooth

Activation Function: Tanh Function

$$\tanh(x) = \frac{1 - \exp(-2x)}{1 + \exp(-2x)}$$



- The tanh (Hyperbolic Tangent) also squashes its inputs into $[-1, 1]$
- The tanh function exhibits point symmetry about the origin

Can a Deep Neural Network approximate a wide range of functions?

Universal Approximation Theorem

Theorem (Universal Approximation Theorem (UAT))

An MLP with a linear output layer and at least one hidden layer with any “squashing” activation function (e.g. logistic sigmoid activation function) can approximate any Borel measurable function from one finite-dimensional space to another with any desired non-zero amount of error, provided that the network is given enough hidden units.

Proof.

See Goodfellow (2016). □

- i.e. A sufficiently large MLP can represent “any” function,
- However, some notes of caution:
 - (i) Theoretical conditions on the UAT don’t always hold,
 - (ii) We are not guaranteed that we will be able to train our MLP to find the true function

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Example 1: Linear ODE

Example 4: (Revisited)

Approaches

- Train neural network to minimize error in differential equation: (This lecture).
 - Raissi-Perdikaris-Karniadakis (2017). *“Physics informed deep learning (part i): Data-driven solutions of nonlinear partial differential equations.”* (Physics Informed Neural Networks (PINNs))
 - Sirignano-Spiliopoulos (2018). *“Deep Galerkin Method (DGM): A deep learning algorithm for solving partial differential equations”*.
 - Gu-Lauriere-Merkel-Payne (2023). *“Deep Learning Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models”*
- Simulate the economy and learn policies that minimize errors along average path:
 - Han, Jentzen, and Weinan (2018). *“Solving high-dimensional partial differential equations using deep learning”*
 - Huang (2022)). *“A probabilistic solution to high-dimensional continuous-time macro-finance models.”*

A General (Quasilinear Parabolic) PDE System

- Consider the (time independent quasilinear parabolic) PDE system:

$$\begin{aligned}\mathcal{L}V(x) &= 0, & x \in \mathcal{X} \\ V(x) &= b(x), & x \in \partial\mathcal{X}\end{aligned}$$

where

- $\mathcal{L}$ is a differential operator. E.g.

$$\mathcal{L}V(x) = \partial_{xx}V(x) + \partial_xV(x)$$

- $\partial\mathcal{X}$ is the boundary of the domain $\mathcal{X}$
- $b(x)$ is the boundary condition

Solution Approach: Idea

- Approximates the function $V(x)$ with a deep neural network $\hat{V}(x; \theta)$, where $\theta \in \mathbb{R}^K$ are the neural network's parameters.
- Goal: chose parameters, θ , to minimise the loss function:

$$\mathcal{E}(\hat{V}) = \left\| \mathcal{L}\hat{V}(x; \theta) \right\|_{\mathcal{X}, \nu_1}^2 + \left\| \hat{V}(x; \theta) - b(x) \right\|_{\mathcal{X}, \nu_2}^2$$

- The norm is given by $\|f(x)\|_{\mathcal{X}, \nu}^2 := \int_{\mathcal{X}} |f(x)|^2 \nu(x) dx$ where $\nu(x)$ is a positive probability density on $x \in \mathcal{X}$.

In practice, we need to approximate the norms by sampling from the state space.

Algorithm

Starting with an initial guess for the parameters θ^0 . At iteration n with guess θ^n :

1. Sample $S^n := \{S^{ni}, S^{nb}\}$ from the state space, where:

- $S^{ni} = \{x_m\}_{m \leq M^e}$ are sample points on **interior** $x_m \in \mathcal{X}$,
- $S^{nb} = \{x_m\}_{m \leq M^b}$ are sample points on **boundary** $x_m \in \partial\mathcal{X}$.

2. Calculate the **weighted average error** on the sample:

$$\mathcal{E}(\theta^n, S^n) = \kappa^i \mathcal{E}^i(\theta^n, S^{ni}) + \kappa^b \mathcal{E}^b(\theta^n, S^{nb}) + \kappa^f \mathcal{E}^f(\theta^n, S^{ni}), \quad \text{where}$$

- $\mathcal{E}^i(\theta^n, S^{ni}) := \frac{1}{M^e} \sum_{m \leq M^e} |\mathcal{L}\hat{V}(x_m)|^2$ is **error in differential equation** on sample S^{ni}
- $\mathcal{E}^b(\theta^n, S^{nb})$ is **error on boundary**
- $\mathcal{E}^f(\theta^n, S^{ni})$ is **penalty for “wrong” shape** (e.g. penalty for non-concavity of V)
- Derivatives are evaluated using **automatic differentiation**.

3. **Update the NN parameters** using a variation on “stochastic” gradient descent:

$$\theta^{n+1} = \theta^n - \alpha_n D_{\theta} \mathcal{E}(\theta^n, S^n)$$

Great in Principle But Implementation is Tricky

- Algorithm has some large advantages for continuous time:
 1. Can deal with high dimensional differential equations,
 2. Calculates derivatives using automatic differentiation (rather than finite difference),
 3. Can sample on random points rather instead of grid.
- However, the algorithm is tricky to implement (despite being easy describe)!
- We need to focus on implementation details (particularly for high dimensions).

Neural Network Q & A

- *Q. Why do we draw new samples each epoch?*
 - Avoid **overfitting** on random samples.
- *Q. Could we use an alternative parametric approx. (e.g. Chebyshev polynomials)?*
 - In principle, yes.
 - However, **automatic derivatives** can easily be calculated for neural networks.
 - and effective **non-linear optimizers** have been developed for neural nets.
- *Q. Does the algorithm solve for global minimum?*
 - Loss calculated by **randomly collected points** so we move around the parameter space.
 - Helps to escape from local minimum trap by SGD (but does not seem to prevent it).

Neural Network Q & A

- *Q. Why do we need shape constraints?*
 - Neural network can find “bad” approximate solutions,
 - Especially likely to find solutions with zero derivative when limited curvature in the problem or dimension is very high,
 - Forcing a shape constraint can prevent this.
- *Q. What if there is a time dimension and an initial condition?*
 - We need to add an additional loss for the initial condition.
- *Q. Why are neural networks better for high dimensional PDEs?*
 - Because we can train them on incrementally of arbitrarily sized samples.
 - That is, we never need to write down a grid.

Algorithm Typical “Hyper-Parameter” Choices

- Number of layers (typically 1 – 5)
- Number of neurons per layer (typically 10 – 100)
- Number of neurons in the output layer (1 per classification class)
- Hidden activation function (I often use the Sigmoid or Tanh)
- Output activation (none; log-odds; soft-max)

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Practical Requirements

- There are technical challenges to working with neural nets (which initially stalled the development)
- Challenge 1: Need to find θ that minimises the loss function
 - Requires an efficient non-linear optimization package
 - Typically uses stochastic gradient descent
- Challenge 2: Need to differentiate the neural network
 - Requires an “automatic differentiation” package that can calculate derivatives efficiently
 - Advances in automatic differentiation have made deep learning feasible (just as they have revolutionised many other areas)

Some Packages

- PyTorch* (what I use)
- TensorFlow
- JAX (what is popular at the moment)
- Scikit learn
- Mxnet
- Caffe
- CNTK
- Flux

References on PyTorch

- PyTorch official tutorials (<https://pytorch.org/tutorials/>) is the best material to start with. The 60-minute blitz provides an overview into constructing deep neural networks with PyTorch.
- PyTorch Docs (<https://pytorch.org/docs/stable/index.html>) serves as the first-hand material for understanding the API, including Torch.nn (<https://pytorch.org/docs/stable/nn.htmz>), Torch.optim (<https://pytorch.org/docs/stable/optim.html>)
- PyTorch Community (<https://discuss.pytorch.org/>) is the “Slack overflow” or “MathWorks” of PyTorch.
- Justin Johnson has a very simple and clean note for PyTorch, which can help you get familiar with all the essentials. (<https://github.com/jcjohnson/pytorch-examples>.)

References on TensorFlow (version 2)

- TensorFlow website: <https://www.tensorflow.org>
- TensorFlow has tutorials (<https://www.tensorflow.org/tutorials/>) and a guide (<https://www.tensorflow.org/guide>) on their website. The Keras guide is a good place to start learning the basic Tensorflow syntax.
- Geron (2019): Has been updated for Keras and TensorFlow 2
- Chollet (2018): This book introduces the basics of deep learning with Keras (but still uses TensorFlow version 1).

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Example 1: Ordinary Differential Equation

- Consider the differential equation:

$$\begin{aligned}y'(x) &= -2xy, & x \in \mathbb{R} \\ y(0) &= 1\end{aligned}$$

- Sample code:
 - Tensorflow: `ode_solver_ex01.py`
 - PyTorch: `ode_solver_ex01_pytorch.py`

Example 2: Ordinary Differential Equation

- Consider the differential equation (steady state diffusion / 1D Poisson)

$$0 = \partial_{xx}g(x) + f(x), \quad x \in (x_L, x_H)$$

- Subject to the Dirichlet boundary conditions:

$$g(x_L) = g_L$$

$$g(x_H) = g_H$$

- And the forcing term:

$$f(x) = e^x$$

- Sample code:
 - Tensorflow: `ode_solver_ex02_loop.py` (manual loop)
 - Tensorflow: `ode_solver_ex02_fit.py` (using Tensorflow fit)
 - PyTorch: `ode_solver_ex02_pytorch.py`

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Collocation Techniques

- Collocation techniques approximate a function $g(x)$ by a linear combination of a finite set of orthogonal basis functions, $\{\varphi_j(x)\}_{1 \leq j \leq N}$:

$$g(x) = \sum_{j=1}^N c_j \varphi_j(x), \quad \text{where } \{c_j\}_{j \leq N} \text{ are coefficients for functions } \{\varphi_j(x)\}_{j \leq N}$$

- Galerkin approach:** Chose coefficients to minimize an appropriate norm of the residual function from the differential equation (e.g. $\|\partial_{xx}g(x) - f(x)\|$).
- Spectral collocation:** Chose coefficients s.t. the differential equation holds at selected grid points and boundary conditions (e.g. $g''(x_i) = f(x_i)$ on grid $\partial_{xx}\{x_i\}_{i \leq I}$).
- This section outlines a simple version of the spectral collocation approach using Chebyshev polynomials.
- Next section uses Neural Networks.

Discretisation of State Space

- Consider a function $g(x)$ on $-1 \leq x \leq 1$.
- Discretise state space: choose $N + 1$ grid points: (The particular points chosen below are called the **Chebyshev points**. See Trefethen (2001), chapter 5 for a discussion of why these points are optimal.)

$$\{x_0, x_1, \dots, x_N\}, \quad x = \cos(j\pi/N), \quad j = 0, 1, \dots, N.$$

- Denote the approximation of g on the grid points by:

$$G := \{G_0, G_1, \dots, G_N\} := \{g(x_0), g(x_1), \dots, g(x_N)\}$$

- We want to approximate the derivatives ($\partial_x g$, $\partial_{xx} g$, ect.) on the grid points

Approximating Derivatives

- Denote the approximation of g on $j \in \{0, 1, \dots, N\}$ by $G := \{G_0, G_1, \dots, G_N\}$.
- Approximating $\partial_x g$ on $j \in \{0, 1, \dots, N\}$ involves two steps:
 1. Find the unique polynomial, p , of degree $\leq N$ that satisfies $p(x_j) = G_j$, $0 \leq j \leq N$. (i.e. the unique polynomial that fits each grid point).
 2. For each $j \in \{0, 1, \dots, N\}$, set $w_j = p'(x_j)$. This is a linear operation and so generates a differentiation matrix, D_N satisfying:

$$\hat{\partial}_x \underline{g} = \underline{w} = D_N \underline{G}, \quad \hat{\partial}_x \underline{g} = \begin{bmatrix} \partial_x g(x)|_{x=x_1} \\ \vdots \\ \partial_x g(x)|_{x=x_N} \end{bmatrix}, \quad \underline{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_N \end{bmatrix}, \quad \underline{G} = \begin{bmatrix} G_1 \\ \vdots \\ G_N \end{bmatrix}$$

- The approximation for $\partial_{xx} g$ on $j \in \{0, 1, \dots, N\}$ is then given by:

$$\hat{\partial}_{xx} \underline{g} = D_N^2 \underline{G}$$

Differentiation Matrix: Example $N = 1$

- Consider the case with 2 grid points. In this case, the interpolation points are:

$$x_0 = \cos(0) = 1, \quad x_1 = \cos(\pi/1) = -1$$

- Let $G = \{G_0, G_1\}$ denote the approximation of $g(x)$ on $\{x_0, x_1\} = \{1, -1\}$.
- Step 1: The interpolating polynomial (satisfying $p(x_0) = G_0$ and $p(x_1) = G_1$) is:

$$p(x) = \frac{1}{2}(1+x)G_0 + \frac{1}{2}(1-x)G_1$$

- Step 2: Taking the derivative gives:

$$p'(x) = \frac{1}{2}G_0 - \frac{1}{2}G_1$$

Differentiation Matrix: Example $N = 1$

- So, the matrix representation for the approximation to $\partial_x g$ is:

$$\hat{\partial}_x g = \begin{bmatrix} \hat{\partial}_x g(x_0) \\ \hat{\partial}_x g(x_1) \end{bmatrix} = \begin{bmatrix} \frac{1}{2}G_0 - \frac{1}{2}G_1 \\ \frac{1}{2}G_0 - \frac{1}{2}G_1 \end{bmatrix} = D_1 \underline{G},$$

where

$$D_1 = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \quad \underline{G} = \begin{bmatrix} G_0 \\ G_1 \end{bmatrix}$$

Differentiation Matrix: Example $N = 2$

- Consider the case with 3 grid points ($N = 2$).
- In this case, the interpolation points:

$$x_0 = 1, \quad x_1 = 0, \quad x_2 = -1$$

- Let $G = \{G_0, G_1, G_2\}$ denote the approximation of $g(x)$ on $\{x_0, x_1, x_2\} = \{1, 0, -1\}$.
- Step 1: The interpolating polynomial (satisfying $p(x_0) = G_0$, $p(x_1) = G_1$, and $p(x_2) = G_2$) is:

$$p(x) = \frac{1}{2}x(1+x)G_0 + (1+x)(1-x)G_1 + \frac{1}{2}x(x-1)G_2$$

- Step 2: Taking the derivative gives:

$$p'(x) = \left(x + \frac{1}{2}\right) G_0 - 2xG_1 + \left(x - \frac{1}{2}\right) G_2$$

Differentiation Matrix: Example $N = 2$

- And so the derivative matrix is:

$$\hat{\partial}_x g = \begin{bmatrix} \hat{\partial}_x g(x_0) \\ \hat{\partial}_x g(x_1) \\ \hat{\partial}_x g(x_2) \end{bmatrix} = \begin{bmatrix} \frac{3}{2}G_0 - 2G_1 + \frac{1}{2}G_2 \\ \frac{1}{2}G_0 - \frac{1}{2}G_2 \\ -\frac{1}{2}G_0 + 2G_1 - \frac{3}{2}G_2 \end{bmatrix} = D_2 \underline{G}$$

where

$$D_2 = \frac{1}{2} \begin{bmatrix} 3 & -4 & 1 \\ 1 & 0 & -1 \\ -1 & 4 & -3 \end{bmatrix}, \quad \underline{G} = \begin{bmatrix} G_0 \\ G_1 \\ G_2 \end{bmatrix}$$

Differentiation Matrix: General

- **Theorem:** Let $\{x_0, x_1, \dots, x_N\}$ denote the Chebyshev points for an $N + 1$ dimensional grid. For each $N \geq 1$, let the rows and columns of the $(N + 1) \times (N + 1)$ Chebyshev spectral differentiation matrix D_N be indexed from 0 to N . The entries of the matrix are:

$$[D_N]_{00} = \frac{2N^2 + 1}{6}, \quad [D_N]_{NN} = -\left(\frac{2N^2 + 1}{6}\right)$$

$$[D_N]_{jj} = \frac{-x_j}{2(1 - x_j^2)}, \quad j = 1, \dots, N - 1$$

$$[D_N]_{ij} = \frac{c_i}{c_j} \frac{(-1)^{i+j}}{(x_i - x_j)}, \quad i \neq j, \quad i, j = 1, \dots, N - 1$$

Differentiation Matrix: General Graphical Depiction

$D_N =$

$\frac{2N^2 + 1}{6}$	$2 \frac{(-1)^j}{1 - x_j}$	$\frac{1}{2}(-1)^N$
$-\frac{1}{2} \frac{(-1)^i}{1 - x_i}$	$\frac{(-1)^{i+j}}{x_i - x_j}$ $\frac{-x_j}{2(1 - x_j^2)}$ $\frac{(-1)^{i+j}}{x_i - x_j}$	$\frac{1}{2} \frac{(-1)^{N+i}}{1 + x_i}$
$-\frac{1}{2}(-1)^N$	$-2 \frac{(-1)^{N+j}}{1 + x_j}$	$-\frac{2N^2 + 1}{6}$

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Example 4: (Revisited)

Example 1 (Revisited)

- Consider the differential equation (steady state diffusion / 1D Poisson)

$$0 = f(x) + \partial_{xx}g(x), \quad x \in (-1, 1)$$

- subject to the Dirichlet boundary conditions:

$$g(x_L) = g_L = 0$$

$$g(x_H) = g_H = 0$$

- and the forcing term:

$$f(x) = e^x$$

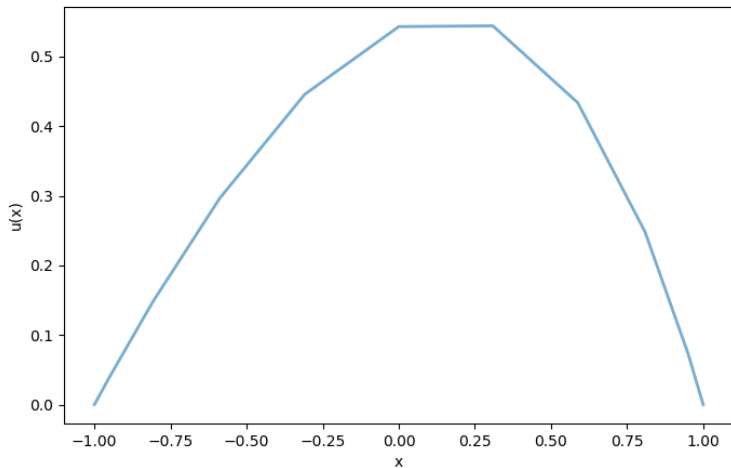
Spectral Approximation to Differential Equation

- Let D_N be the N-grid point spectral approximation to ∂_x and let $D_N^2 = D_N D_N$ be the N-grid point spectral approximation to $\partial_x x$. Then, the DE is:

$$\begin{aligned}G_1 &= 0, \quad j = 1 \\[D_N^2]_{(j,\cdot)} \cdot G &= f(x_j), \quad j = 2, \dots, N-1 \\G_N &= 0, \quad j = N\end{aligned}$$

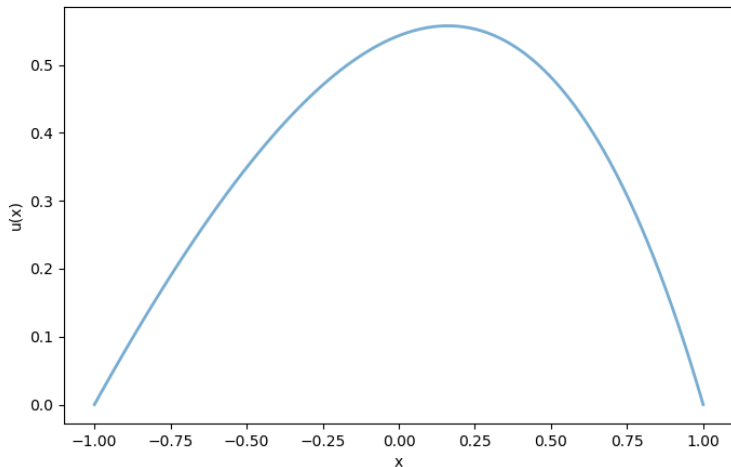
- So, for this problem, the outermost rows and columns in D_N^2 play no role and so can be excluded.

Sample Code (No Smoothing)



Julia/SM_Pois1DEqnSolver.jl

Sample Code (Polynomial Smoothing)



Julia/SM_Pois1DEqnSolver.jl

Problems on Other Domains

- What if we want to solve a problem on $[a, b]$ not on $x \in [-1, 1]$.
- **Option 1:** Find new optimal Chebyshev points:

$$x_i = a + \frac{1}{2}(b - a)(1 + \cos(\pi(1 - z_i))), \quad \text{for } i = 0, 1, \dots, N$$

where z_i are evenly spaced in the unit interval: $z_i = i/(N)$, for $i = 0, 1, \dots, N$. Then, find the matrices for those points.

- **Option 2:** Use change of variables in the problem to get domain into $[-1, 1]$

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Optional Appendix: Spectral Method

General Principles

Example 1: Linear ODE

Example 4: (Revisited)

Example 1 (Revisited)

- Consider the differential equation (steady state diffusion / 1D Poisson)

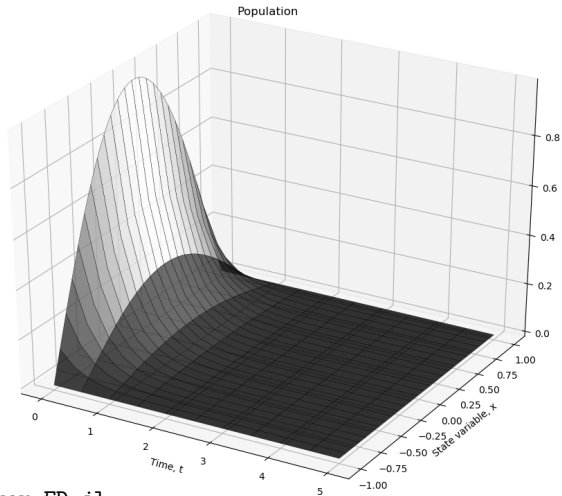
$$0 = f(x) + \partial_{xx}g(x), \quad x \in (-1, 1)$$

- subject to the Dirichlet boundary conditions:

$$g(x_L) = g_L = 0$$

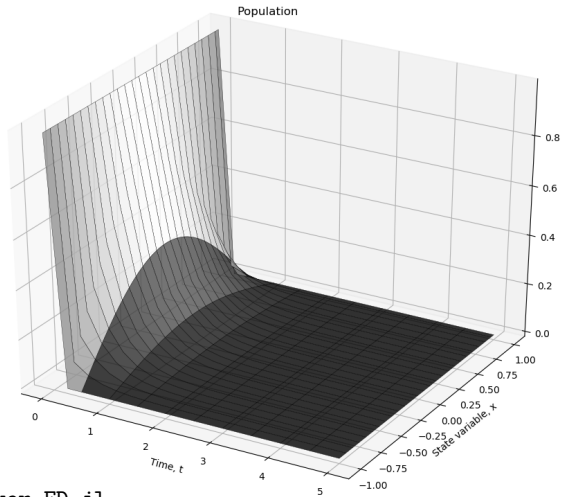
$$g(x_H) = g_H = 0$$

Sample Code ($g(x, 0) = \cos(x\pi/2)$, $f(x) = 0$)



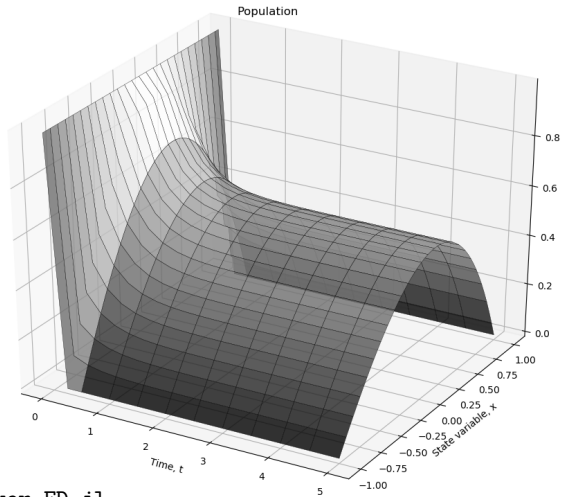
Julia/FD_Ex04_HeatEqnSolver_FD.jl

Sample Code ($g(x, 0) = 1, f(x) = 0$)



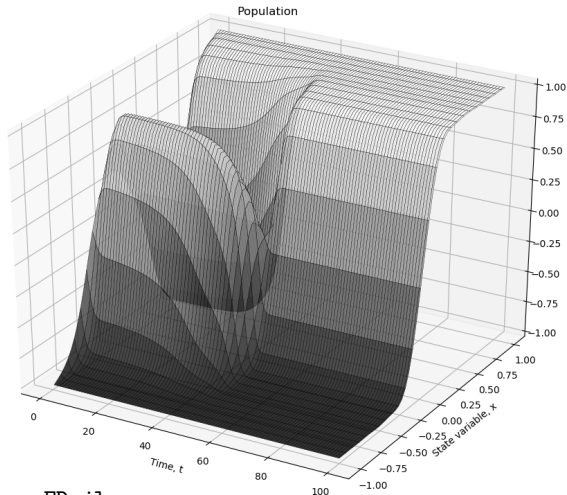
Julia/FD_Ex04_HeatEqnSolver_FD.jl

Sample Code ($g(x, 0) = 1, f(x) = 0$)



Julia/FD_Ex04_HeatEqnSolver_FD.jl

Sample Code ($g(x, 0) = .53x + .47 \sin(-1.5\pi x)$, $f(x) = x - x^3$)



Julia/FD_Ex04_HeatEqnSolver_FD.jl

Q: What are some potential problems with this technique?

Q: Could we use this technique for a differential equation with 100 dimensions?

Q/A

Q: What are some potential problems with this technique?

- **A.** The method can better handle curvature but still needs a grid.
- We need to move towards random sampling (i.e. the Galerkin method)

Q: Could we use this technique for a differential equation with 100 dimensions?

- **A.** No chance!
- If we have N grid points in very dimension, then total matrix dimension is N^{100} .
- Need to go to deep learning to solve high dimensional problems.